EPC data: Technical documentation

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Introduction

This document describes the Energy Performance Certificate (EPC) data collected for SERL participants, stored in the file *serl_epc_data_edition07.csv*. The data contains 103 columns and 7132 rows (one row per participant with available EPC data). The number of participants included is likely to increase from one edition to the next as new EPCs are lodged for households that previously did not have one. EPC data was only available for England and Wales until the 4th Observatory edition; since then EPC data has been available for England, Wales and Scotland.

This document lists the EPC variables available along with basic information about the values for each variable such as number of unique values and statistics for numerical variables. There are some differences between the English and Welsh EPCs versus the Scottish EPCs, both in terms of the variables available and the structure of the data. In particular, for the Scottish data if there are multiple types of an element in a dwelling then these are recorded in the relevant column and separated by "|" [ASCII (decimal) 124], whereas only one type is recorded per element for England and Wales. A guide to the variables in the English and Welsh data is available here, and for Scottish data is available here.

Data were collected with the Domestic Energy Performance Certificates API using the house number and postcode (details for England and Wales here). Details about downloading the Scottish data are available here. Where more than one EPC is registered for an address the most recent is provided.

The data have not been modified from the original source except for the removal of address data (replaced with our PUPRN (a unique identifier) used in the other SERL datasets), replacement of empty strings with NA values, and removal of the Scottish variable 'addendumText' as this did not contain data for any participants. Note that data quality analysis has not been performed for this dataset.

Data summary

Table 1 lists all variables currently available in the SERL EPC dataset. The number of unique values is given, alongside the R data class and an example value from the dataset. The countries for which each variable is available is also given, with 'All' meaning the variable is available in both the English and Welsh and Scottish EPC data.

For variables with fewer than 10 unique values in the EPC dataset, Table 2 shows the number of records with each value and the percent with this value (or non-value in the case of N/A or 'NO DATA!' etc.). We also include PUPRN to show the number of records.

Table 1: All EPC variables, the number of unique values found for each variable, the variable (R) class, and an example from the dataset. In some cases examples are invented (for statistical disclosure control) as the count for unique reads is very low; in these instances examples take the same format as the real data.

variable	*n unique values*	*class*	*example*	*countries available*
PUPRN	7,132	character	1ABC2DE3	All
builtForm	7	character	Detached	All
co2EmissCurrPerFloorArea	151	numeric	24	All
co2EmissionsCurrent	133	numeric	1	All
co2EmissionsPotential	129	numeric	1.4	All
constituency	839	character	E14000734	All
constructionAgeBand	29	character	1976-1983	All
currentEnergyEfficiency	97	integer	44	All
currentEnergyRating	7	character	B	All
energyConsumptionCurrent	574	integer	327	All
energyConsumptionPotential	467	integer	184	All
energyTariff	11	character	Single	All
environmentImpactCurrent	99	integer	44	All
environmentImpactPotential	92	integer	95	All
extensionCount	6	integer	2	All
fixedLightingOutletsCount	68	integer	18	All

variable	n unique values	class	example	countries available
flatStoreyCount	11	integer	2	England and Wales
flatTopStorey	16	character	N	All
floorDescription	123	character	Solid, no insulation (assumed)	All
floorEnergyEff	9	character	Very Good	All
floorEnvEff	8	character	Good	All
floorHeight	198	numeric	2.45	All
floorLevel	43	character	03	All
glazedArea	12	character	Normal	All
glazedType	11	character	triple glazing	All
heatingCostCurrent	2,047	integer	682	All
heatingCostPotential	1,647	integer	456	All
heatLossCorridor	5	character	heated corridor	England and Wales
hotWaterCostCurrent	573	integer	127	All
hotWaterCostPotential	398	integer	201	All
hotwaterDescription	42	character	Community scheme	All
hotWaterEnergyEff	6	character	Very Good	All
hotWaterEnvEff	6	character	Poor	All
inspectionDate	3,494	character	12/04/2022	All
lightingCostCurrent	369	integer	100	All
lightingCostPotential	313	integer	27	All
lightingDescription	111	character	Low energy lighting in 35% of fixed outlets	All
lightingEnergyEff	6	character	Average	All
lightingEnvEff	6	character	Very Good	All
localAuthority	345	character	E06000016	England and Wales

variable	n unique values	class	example	countries available
lodgementDate	3,613	character		All
lodgementDatetime	6,483	character	02/10/2010 14:51	England and Wales
lowEnergyFixedLightCount	43	integer	6	All
lowEnergyLighting	99	integer	55	All
mainFuel	33	character	oil (not community)	All
mainheatcEnergyEff	8	character	Poor	All
mainheatcEnvEff	8	character	Very Good	All
mainheatcontDescription	54	character	Room thermostat only	All
mainheatDescription	75	character	Room heaters, mains gas	All
mainheatEnergyEff	10	character	Very Good	All
mainheatEnvEff	9	character	Average	All
mainHeatingControls	48	integer	2699	All
mainsGasFlag	3	character	N	All
mechanicalVentilation	5	character	natural	All
multiGlazeProportion	85	integer	70	All
numberHabitableRooms	17	integer	2	All
numberHeatedRooms	17	integer	5	All
numberOpenFireplaces	10	integer	0	All
photoSupply	63	character	Array: Roof Area: 0%; Connection: not applicable (FGHRS or no PV); |	All
potentialEnergyEfficiency	92	integer	87	All
potentialEnergyRating	7	character	F	All
propertyType	5	character	Maisonette	All
roofDescription	234	character	Pitched, 270 mm loft insulation	All
roofEnergyEff	41	character	Poor	All

variable	*n unique values*	*class*	*example*	*countries available*
roofEnvEff	41	character	Poor	All
secondheatDescription	27	character	Room heaters, electric	All
sheatingEnergyEff	2	character		All
sheatingEnvEff	2	character	N/A	All
solarWaterHeatingFlag	4	character	Y	All
tenure	12	character	rented (social)	All
totalFloorArea	1,315	numeric	44	All
transactionType	17	character	rental	All
unheatedCorridorLength	397	numeric	6	All
wallsDescription	119	character	Timber frame, as built, partial insulation (assumed)	England and Wales
wallsEnergyEff	6	character	Very Poor	England and Wales
wallsEnvEff	6	character	Poor	England and Wales
windowsDescription	32	character	High performance glazing	All
windowsEnergyEff	6	character	Very Poor	All
windowsEnvEff	6	character	Poor	All
windTurbineCount	4	integer	0	All
3YrEnergyCostCurrent	523	integer		Scotland
3YrEnergySavingPotential	386	integer	0	Scotland
airTightnessDescription	9	character		Scotland
airTightnessEnergyEff	3	character		Scotland
airTightnessEnvEff	3	character		Scotland
alternativeImprovements	21	character	Description: External insulation with cavity wall insulation |	Scotland
currentEnvironmentalRating	8	character	E	Scotland

variable	n unique values	class	example	countries available
dataZone	571	character		Scotland
heatLossCorridoor	4	character	unheated corridor	Scotland
impactCavityWallInsulation	107	integer	0	Scotland
impactLoftInsulation	194	integer	0	Scotland
impactSolidWallInsulation	140	integer	0	Scotland
improvements	598	character		Scotland
lzcEnergySources	19	character	Description: Biomass secondary heating |	Scotland
mainHeatingCategory	6	character	boiler with radiators or underfloor heating	Scotland
potentialEnvironmentalRating	6	character	B	Scotland
spaceHeatingDemand	611	integer		Scotland
typeOfAssessment	3	character	RdSAP, existing dwelling	Scotland
wallDescription	106	character	System built, as built, no insulation (assumed)	Scotland
wallEnergyEff	38	character	Poor | Good	Scotland
wallEnvEff	38	character	Good | Good	Scotland
waterHeatingDemand	498	integer		Scotland
epcVersion	2	character	England and Wales	All

Table 2: The number and percent of each value found in the dataset for each variable with fewer than 10 unique values found. Note that number is rounded down to the nearest 10 for statistical disclosure control and values with less than 10 participants have been removed. The percent is the rounded number as a percent of total (so might not sum to 100%).

variable	value	number (rounded)	percent
builtForm	Detached	1,970	27.6
	Enclosed End-Terrace	80	1.1
	Enclosed Mid-Terrace	50	0.7
	End-Terrace	870	12.2

variable	value	number (rounded)	percent
	Mid-Terrace	1,740	24.4
	NO DATA!	60	0.8
	Semi-Detached	2,340	32.8
currentEnergyRating	A	10	0.1
	B	440	6.2
	C	2,440	34.2
	D	3,010	42.2
	E	990	13.9
	F	170	2.4
	G	40	0.6
extensionCount		310	4.3
	0	4,120	57.8
	1	1,920	26.9
	2	590	8.3
	3	130	1.8
	4	40	0.6
floorEnergyEff		3,850	54.0
	Good	50	0.7
	N/A	510	7.2
	N/A | N/A	70	1.0
	NO DATA!	2,440	34.2
	Very Good	170	2.4
floorEnvEff		6,300	88.3
	Good	50	0.7
	N/A	510	7.2
	N/A | N/A	70	1.0

variable	value	number (rounded)	percent
	Very Good	170	2.4
heatLossCorridor		1,670	23.4
	NO DATA!	4,370	61.3
	heated corridor	150	2.1
	no corridor	400	5.6
	unheated corridor	520	7.3
hotWaterEnergyEff		10	0.1
	Average	900	12.6
	Good	5,130	71.9
	Poor	400	5.6
	Very Good	360	5.0
	Very Poor	300	4.2
hotWaterEnvEff		10	0.1
	Average	740	10.4
	Good	5,290	74.2
	Poor	570	8.0
	Very Good	350	4.9
	Very Poor	150	2.1
lightingEnergyEff	Average	1,030	14.4
	Good	1,300	18.2
	Poor	600	8.4
	Very Good	3,280	46.0
	Very Poor	880	12.3
lightingEnvEff	Average	1,030	14.4
	Good	1,300	18.2
	Poor	600	8.4

variable	value	number (rounded)	percent
	Very Good	3,280	46.0
	Very Poor	890	12.5
mainheatcEnergyEff		10	0.1
	Average	2,070	29.0
	Good	4,140	58.0
	Poor	380	5.3
	Very Good	200	2.8
	Very Poor	310	4.3
mainheatcEnvEff		10	0.1
	Average	2,070	29.0
	Good	4,140	58.0
	Poor	380	5.3
	Very Good	200	2.8
	Very Poor	310	4.3
mainheatEnvEff		10	0.1
	Average	300	4.2
	Good	6,040	84.7
	Poor	180	2.5
	Very Good	340	4.8
	Very Poor	240	3.4
mainsGasFlag		830	11.6
	N	720	10.1
	Y	5,570	78.1
mechanicalVentilation	NO DATA!	310	4.3
	mechanical, extract only	30	0.4
	mechanical, supply and extract	10	0.1

variable	value	number (rounded)	percent
	natural	6,760	94.8
potentialEnergyRating	A	210	2.9
	B	3,640	51.0
	C	2,700	37.9
	D	450	6.3
	E	90	1.3
	F	10	0.1
propertyType	Bungalow	940	13.2
	Flat	1,260	17.7
	House	4,800	67.3
	Maisonette	110	1.5
sheatingEnergyEff		6,530	91.6
	N/A	600	8.4
sheatingEnvEff		6,530	91.6
	N/A	600	8.4
solarWaterHeatingFlag		1,930	27.1
	N	5,140	72.1
	Y	50	0.7
wallsEnergyEff		620	8.7
	Average	800	11.2
	Good	2,860	40.1
	Poor	1,050	14.7
	Very Good	280	3.9
	Very Poor	1,500	21.0
wallsEnvEff		620	8.7
	Average	800	11.2

variable	value	number (rounded)	percent
	Good	2,860	40.1
	Poor	1,050	14.7
	Very Good	280	3.9
	Very Poor	1,500	21.0
windowsEnergyEff	Average	4,000	56.1
	Good	2,320	32.5
	Poor	270	3.8
	Very Good	210	2.9
	Very Poor	290	4.1
windowsEnvEff	Average	4,000	56.1
	Good	2,320	32.5
	Poor	270	3.8
	Very Good	210	2.9
	Very Poor	290	4.1
windTurbineCount		180	2.5
	0	6,940	97.3
airTightnessDescription		7,120	99.8
airTightnessEnergyEff		7,120	99.8
airTightnessEnvEff		7,120	99.8
currentEnvironmentalRating		6,510	91.3
	B	50	0.7
	C	210	2.9
	D	200	2.8
	E	100	1.4
	F	30	0.4
heatLossCorridoor		6,940	97.3

variable	value	number (rounded)	percent
	no corridor	60	0.8
	unheated corridor	120	1.7
mainHeatingCategory		6,510	91.3
	boiler with radiators or underfloor heating	570	8.0
	electric storage heaters	10	0.1
	heat pump with radiators or underfloor heating	10	0.1
potentialEnvironmentalRating		6,510	91.3
	A	30	0.4
	B	300	4.2
	C	230	3.2
	D	40	0.6
typeOfAssessment		6,510	91.3
	RdSAP, existing dwelling	610	8.6
epcVersion	England and Wales	6,510	91.3
	Scotland	610	8.6